Vejay Suriyaa Senguttuvan

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EDUCATION

University of Southern California (USC), Los Angeles, CA Aug 2025 – May 2027 (Expected)
Master of Science in Electrical and Computer Engineering
Relevant Coursework: Diagnosis and Design of Reliable Digital Systems (DFT), VLSI System Design, MOS VLSI Circuit Design, Computer Systems Organization, Integrated Memory Devices and Technologies, Reinforcement Learning and LLMs

SRM Institute of Science and Technology, Chennai, India Nov 2020 – May 2024
Bachelor of Technology in Electronics and Computer Engineering

TECHNICAL SKILLS

Verification and Test: Test planning, golden/reference models (Python), self-checking tests, test-first (model-before-RTL) flow, fault simulation, ATPG (D-Algorithm, PODEM, SCOAP), scan insertion, BIST (LFSR/MISR), stuck-at and transition fault models, memory test
HDL and Programming: Verilog, SystemVerilog (RTL and testbench fundamentals), C++, Python (NumPy), Tcl (Innovus scripting), MATLAB, Bash, Make; Linux, Git/GitHub
AI / ML: Reinforcement learning (MDPs, value/policy methods, policy gradient), LLM fundamentals (transformers, RL fine-tuning), RL for EDA, in-memory computing for AI hardware (RRAM crossbars, SRAM compute-in-memory)
AI-Native Development: Agentic coding with Claude Code (spec-driven builds, CLAUDE.md context engineering, autonomous overnight builds, AI code review in GitHub), agent workflow design, prompt engineering, LLM workflow automation
EDA Tools: Cadence Xcelium (RTL simulation), SimVision (waveform debug), Innovus (place and route, ecoRoute, verify_drc), Virtuoso (schematic and layout), KLayout (DRC), mflowgen, HSPICE, LTspice, ModelSim; SkyWater SKY130 PDK

PROJECTS

Camera ISP Pipeline, RTL and Verification | Tiny Tapeout TTSKY26d (SkyWater SKY130)
Python, NumPy, Verilog, Cadence Xcelium · shuttle deadline Nov 30, 2026
- Wrote the test plan, a fixed-point Python golden model for 2 ISP stages (black-level correction, white balance with Q2.6 gains) and self-checking unit tests before any RTL existed (test-first), the reference for RTL verification
- Next: Verilog RTL for the first stages, verified in Cadence Xcelium against the model's vectors, for the Nov 30, 2026 shuttle

RL Agent for Post-Route DRC Closure (AI for EDA)
Python, Tcl, Cadence Innovus, mflowgen, KLayout, SkyWater SKY130 · EE 599 team project
- Own the measurement side of a 4-person team building an RL agent that fixes post-route DRC violations with minimal layout disruption; brought up the mflowgen + Innovus + SKY130 flow on the USC server and froze the IBEX (RISC-V, 12.4k cells) benchmarks
- Wrote the DRC report parser (Innovus verify_drc, KLayout) and a DEF-diff disruption metric (nets rerouted, wirelength, vias, cells moved), both reward inputs; found KLayout misses 82% of AES violations (all 47 shorts), so the reward uses Innovus counts
- Built the native ecoRoute baseline the agent must beat and measured it on 5 frozen designs (AES 57 to 48 violations, a 15.8% reduction, in 111 s touching 1.65% of nets; IBEX 838 to 0 touching 12% of nets); fixed a router setting that made ecoRoute a no-op on onepass runs

Combinational ATPG Engine, Test Point Insertion, BIST Simulator and Scan Insertion
C++17, Python, ISCAS-85 · EE 658 course project · github.com/vejaysuriyaa-07/atpg-tpi-ppa
- Implemented a combinational ATPG engine in C++17 (D-Algorithm with D/J-frontier tracking and SCOAP heuristics); 97.37 to 100% stuck-at fault coverage across 6 ISCAS-85 circuits using checkpoint-collapsed fault lists
- Evaluated test-point insertion with 1,000 LFSR patterns per seed across 5 seeds; raised c432 coverage from 98.36% to 99.87% with 4 control points at 6.9% estimated area overhead
- Built fault simulation, Galois LFSR/MISR BIST and full-scan insertion for sequential .bench netlists; 22 GoogleTest cases

RTL Live, VS Code Extension (open source)
VS Code Marketplace v0.5.1 · github.com/vejaysuriyaa-07/rtl-live
- Built and published a VS Code extension that renders a live Verilog schematic in the sidebar on every edit, with 2 synthesized views
- Delivered with an agentic AI coding workflow (Claude Code): spec, acceptance criteria, overnight autonomous build, release in a week

EXPERIENCE

Course Producer, EE 658 Diagnosis and Design of Reliable Digital Systems (graduate DFT), USC Aug 2026 – Present
- Selected by Prof. Moe Tabar after earning an A; grade homework, projects and exams on fault modeling, ATPG, fault simulation and BIST
- Hold weekly office hours on test generation, fault simulation and testability problems; set assignment deadlines and proctor exams

Research Assistant, In-Memory Computing and VLSI Jan 2025 – Jul 2025
Prof. Mohanbabu A.'s lab, SRM IST Chennai (part-time)
- Simulated RRAM crossbar arrays in HSPICE and LTspice for analog in-memory computing (IMC) matrix-vector multiplication; worked on SRAM compute-in-memory circuits in Cadence Virtuoso with Verilog/ModelSim for the digital side
- Modeled in Python and MATLAB how device variation and limited precision degrade neural-network accuracy on IMC hardware

PUBLICATIONS

- D. Manoj Kumar, R. Arthi, M. Gokulakrishnan, T. Praveen Kumar, **S. Vejaysuriyaa**, "A Food Wastage Reduction App based on Django Python Application," 3rd Int. Conf. on Applied Artificial Intelligence and Computing (ICAAIC), IEEE, June 2024. IEEE Xplore

LEADERSHIP

ECE Senator, Viterbi Graduate Student Association, USC (Aug – Dec 2025) · **Head of Photography, Camogenics**, SRM (2022 – 2024)